

Task-Appropriate Hybrid Architectures: A Unified Theory of Gain Across Metrics, Architectures, and Tasks

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Abstract

We present a unified theory of gain for hybrid systems across three independent dimensions: (1) metric type—linear (accuracy, bit-accuracy, token-accuracy) versus non-linear (F_1); (2) architecture type—bilinear Kronecker versus dense multi-layer perceptron; and (3) task structure—random linear versus compositional (grid-aligned). For each dimension we determine *when* which structure wins, with statistical evidence from 20–50 independent seeds. We extend the linear metric formula of our earlier work to a per-class identity that is exact for macro F_1 (machine-precision error on 20 NER configurations), and we show that the Kronecker architecture is $8\times$ more parameter-efficient on simple compositional tasks and statistically superior on image-like compositional tasks (50 seeds, $p < 0.0001$, Cohen’s $d = -1.37$). We conclude that hybrid system design can be made principled: formula-driven for metrics, structure-driven for tasks.

1 Introduction

Hybrid systems that combine symbolic reasoning with neural learning have become a central theme of modern AI research. Yet the question of *when* a hybrid system outperforms its neural-only baseline remains largely heuristic. Prior work has proposed empirical guidelines, but no unified predictive framework exists.

Our earlier work [?] established a closed-form formula for the gain of a hybrid symbolic-neural system over a neural-only baseline, valid for linear metrics:

$$\text{gain} = \text{cov}_{\text{sym}} \cdot (\text{acc}_s - \text{acc}_n), \quad (1)$$

where cov_{sym} is the symbolic arm’s coverage and $\text{acc}_s, \text{acc}_n$ are the symbolic and neural accuracies. This formula was validated on 12 languages, 4 task types, and 2 metrics with mean prediction error 0.002.

The present paper addresses three open questions that the linear formula does not cover:

1. **Metric scope.** Does the formula generalise to non-linear metrics such as F_1 ?
2. **Architecture choice.** When is a bilinear Kronecker architecture preferable to a standard dense MLP?
3. **Task structure.** How does the gain depend on whether the task is random-linear or compositional?

Contributions.

1. We extend the linear formula to a per-class identity that is *exact* for macro F_1 (Section ??).
2. We show that the Kronecker architecture is $8\times$ more parameter-efficient on compositional tasks, and that hierarchical Kronecker statistically dominates dense MLPs on image-like compositional tasks (Section ??).
3. We provide a decision tree that maps task type to architecture choice, backed by 20–50-seed paired t -tests.
4. We document negative results honestly (Sections ?? and ??).

2 Theory

2.1 Linear Metric Formula

Let $s(x) \in \{0, 1, \perp\}$ denote the symbolic arm’s output where $\perp$ is abstention, and $n(x) \in \{0, 1\}$ the neural arm’s output. The hybrid rule is $h(x) = s(x)$ if $s(x) \neq \perp$, else $n(x)$.

Define coverage $\text{cov} = \Pr[s(x) \neq \perp]$ and $\text{acc}_n^\perp = \Pr[n(x) = y \mid s(x) = \perp]$. For any metric M that is *linear* in the per-item correctness indicator, the hybrid’s value decomposes exactly as:

$$M_h = \text{cov} \cdot M_s + (1 - \text{cov}) \cdot M_n^\perp. \quad (2)$$

Subtracting $M_n = \text{cov} \cdot M_n + (1 - \text{cov}) \cdot M_n$, we obtain the gain formula:

$$\text{gain} = \text{cov} \cdot (M_s - M_n) + (1 - \text{cov}) \cdot (M_n^\perp - M_n). \quad (3)$$

The simple predictive form drops the second term when $M_n^\perp \approx M_n$; the residual is bounded by $\varepsilon = (1 - \text{cov}) \cdot |M_n^\perp - M_n|$.

2.2 F1 Per-Class Formula

Macro F_1 is not linear in the per-item indicator, so the identity of Equation ?? fails. We derive an exact per-class decomposition instead.

Let K be the number of classes. For class c , define the *positive mass* of a predictor as

$$D^c = 2 \text{TP}^c + \text{FP}^c + \text{FN}^c. \quad (4)$$

The classwise F_1 is $F_1^c = 2 \text{TP}^c / D^c$.

For the hybrid system, both TP and D decompose across the answered and abstained subsets:

$$\text{TP}_h^c = \text{TP}_s^c + \text{TP}_n^c, \quad (5)$$

$$D_h^c = D_s^c + D_n^c. \quad (6)$$

Therefore:

$$F_{1h}^c = \frac{2(\text{TP}_s^c + \text{TP}_n^c)}{D_s^c + D_n^c} \quad (7)$$

$$= \frac{D_s^c \cdot F_{1s}^c + D_n^c \cdot F_{1n}^c}{D_s^c + D_n^c} \quad (8)$$

$$= w^c \cdot F_{1s}^c + (1 - w^c) \cdot F_{1n}^c, \quad (9)$$

where the per-class weight is

$$w^c = \frac{D_s^c}{D_s^c + D_n^c}. \quad (10)$$

Averaging over classes gives the macro identity:

$$F_{1,h} = \frac{1}{K} \sum_{c=1}^K [w^c F_{1s}^c + (1 - w^c) F_{1n}^c]. \quad (11)$$

Relation to linear formula. For a *linear* metric, one can show that $\sum_c D^c = 2N$ holds identically for any prediction, so the aggregate weight collapses to $w = \text{cov.}$ For macro F_1 , however, the weights w^c are per-class and do *not* collapse: class imbalance on the abstained subset breaks the identity. This is precisely why the linear formula fails for F_1 and the per-class formula succeeds.

2.3 Kronecker Task-Appropriateness

A bilinear Kronecker layer computes $Y = AXB$ for $X \in \mathbb{R}^{n \times n}$. Flattened, this is $Y_{\text{vec}} = (B^\top \otimes A)X_{\text{vec}}$, a linear operator of size $n^2 \times n^2$ represented by only $2n^2$ trainable parameters. The operator’s rank is $\text{rank}(A) \cdot \text{rank}(B) = n^2$ (full rank), but its degrees of freedom remain $2n^2$.

This structural constraint is a *feature* when the task’s own structure aligns with the tensor decomposition. We formalise this intuition as:

Task-Appropriateness Hypothesis. Kronecker architectures are parameter-efficient when the task’s functional dependencies respect the tensor factorisation $X \mapsto AXB$. Conversely, on tasks whose target is a generic (non-factorisable) linear map, the same constraint becomes a bottleneck.

We test this hypothesis across three task types in Section ??.

3 Experiments

3.1 Linear Metric Validation

Equation ?? was validated on 12 datasets spanning 8 language families, 4 writing systems (Latin, Han, Cyrillic, Arabic), and 4 task types (binary dependency verification, multiclass, multi-label, NER). Table ?? summarises the aggregate error.

Table 1: Linear metric formula validation. Mean prediction error across all configurations.

Dimension	Coverage
Languages	12 (8 families)
Writing systems	4 (Latin, Han, Cyrillic, Arabic)
Task types	4 (binary, multiclass, multi-label, NER)
Metrics	2 (accuracy, bit-accuracy)
Mean error	0.002
Max error	0.008

Under symbolic noise (0–50%), the formula’s prediction error remains invariant at $\varepsilon = 0.0066$, indicating that the linear identity captures a structural property of the hybrid ensemble.

3.2 F1 Per-Class Validation

The linear formula fails on F_1 because F_1 involves a harmonic mean and is not linear in the per-item correctness indicator. We tested this on 20 NER configurations (5 seeds $\times$ 4 hidden ratios).

The per-class formula achieves machine-precision error on all 20 configurations, confirming that the extension is exact for macro F_1 .

Table 2: F_1 formula comparison on 20 NER configurations. The linear formula uses cov as weight; the per-class formula uses $w_c = D_s^c / (D_s^c + D_n^c)$.

Formula	Mean ε	Max ε
Linear (cov-weighted)	0.0188	0.0515
Per-class (w_c -weighted)	0.000000	0.000000

3.3 Kronecker Task-Appropriateness Results

We tested the task-appropriateness hypothesis across three task types: a random linear teacher (synthetic), a simple compositional task (block powers), and an image-like compositional task (2D grid with spatial correlation). Table ?? summarises the winner in each case.

Table 3: Winner by task type. Parameter counts in parentheses. Statistical tests: paired t -test on matched seeds.

Task type	Winner	Test Acc	Efficiency
Random linear	Dense MLP	—	baseline
Simple compositional	Flat Kronecker	0.8007	8× fewer
Image compositional	Hier. Kronecker	0.8360	34× more

Simple compositional. 20 seeds. Flat Kronecker (2,564 params) reaches 0.8007 ± 0.0202 versus Dense MLP (20,868 params) at 0.7982 ± 0.0190 ; the difference is not significant ($p = 0.52$). Hierarchical Kronecker (87,044 params) reaches 0.7872 and is *significantly worse* than flat ($p = 0.0008$, Cohen’s $d = -0.75$).

Image compositional. 50 seeds. Hierarchical Kronecker reaches 0.8360 ± 0.0217 versus Dense MLP at 0.8190 ± 0.0198 (paired t -test: $p < 0.0001$, Cohen’s $d = -1.37$). Flat Kronecker (2,564 params) reaches 0.8267 and is significantly better than Dense MLP ($p = 0.0003$, $d = +0.51$), demonstrating 8× parameter efficiency with a small absolute accuracy advantage.

The contrast between the two compositional tasks—simple blocks versus image-like grids with spatial correlation—demonstrates that Kronecker’s advantage depends on the *degree* of structural alignment between the task and the tensor factorisation.

3.4 Negative Results

We document two negative results that bound the Kronecker claim.

Parameter-matched comparison. On the random linear task, when both architectures are given the same parameter budget ($\sim 21,764$), Dense MLP reaches 0.900 while Flat Kronecker reaches 0.257 at depth $K = 168$. Deep flat chains collapse: repeated bilinear + SiLU without normalisation destroys information. Hierarchical Kronecker partially avoids collapse (0.816) but still loses to Dense MLP.

LayerNorm + residual. Adding LayerNorm and residual connections to flat Kronecker recovers from 0.402 to 0.584, but does not close the gap to hierarchical (0.817) or dense (0.916). We conclude that the flat chain’s limitation is not purely a normalisation problem.

These results establish that the Kronecker advantage is *task-conditional*, not universal.

4 Discussion

4.1 When to Use Which

Our experiments yield a simple decision rule:

1. **Metric choice.** Use linear metrics (accuracy, bit-accuracy) when possible; the gain formula is closed-form (Equation ??). For macro F_1 , use the per-class weighted identity.
2. **Architecture choice.** Match the architecture to the task structure. Random linear tasks favour Dense MLP; block-structured tasks favour Flat Kronecker; image-like tasks with spatial correlation favour Hierarchical Kronecker.
3. **Hierarchy.** Do not deploy hierarchical Kronecker on tasks without spatial structure: it costs $34\times$ more parameters and underperforms.

4.2 Why Does the F1 Formula Require Per-Class Weights?

For linear metrics, the weight collapses to coverage: $\sum_c D_c = 2N$ identically, so $w = \text{cov}$. For macro F_1 , per-class weights $w_c = D_s^c / (D_s^c + D_n^c)$ differ across classes and from cov; the correct aggregation is necessarily per-class.

4.3 The Structural Alignment Principle

The Kronecker results suggest a broader principle: *a parameter budget is most efficiently spent when its implicit constraints align with the task’s dependencies*. A bilinear layer constrains the operator to the form $B^\top \otimes A$. When the task’s target respects this form, the constraint is free; when it does not, the constraint becomes a bottleneck.

5 Limitations and Future Work

Limitations. The Kronecker task-appropriateness study covers three synthetic task types; generalisation to real-world image or language data is an open question. The per-class F_1 identity is proven for macro F_1 ; weighted and micro variants are not addressed. Our neural baselines are small models trained from scratch; we do not compare against pretrained language models. Statistical power is adequate ($n = 20\text{--}50$) but not exhaustive.

Future work. (1) Test the Kronecker hypothesis on natural images and text. (2) Derive per-class identities for weighted F_1 , precision, and recall. (3) Extend the hybrid formula to pretrained large models. (4) Investigate whether task-appropriateness can be predicted from data statistics alone.

6 Conclusion

Hybrid systems do not win unconditionally. Their gain depends on three independent dimensions: metric type (linear vs non-linear), architecture type (Kronecker vs dense), and task structure (random vs compositional). We provide a formula for each dimension, statistical evidence for the architecture choice, and an honest account of when the geometric architecture loses. Together, these results suggest that hybrid system design can be made principled rather than heuristic.

References

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